

Deviations from Mendelism

Worksheet · PU-II Biology, Chapter 4

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Worksheet – Deviations from Mendelism

15 questions · answer key on the last page

- Q1.** Incomplete dominance was discovered by which scientist, working on *Mirabilis jalapa*?
- (a) Gregor Mendel
 - (b) Carl Correns
 - (c) Karl Landsteiner
 - (d) Thomas Hunt Morgan
- Q2.** In incomplete dominance, the F1 phenotype:
- (a) Resembles one parent completely
 - (b) Is an intermediate/mixture of both parents, resembling neither parent
 - (c) Always resembles the mother
 - (d) Shows both parental phenotypes simultaneously, fully expressed
- Q3.** In *Mirabilis jalapa*, a cross between red-flowered (RR) and white-flowered (rr) plants gives an F1 that is:
- (a) Red
 - (b) White
 - (c) Pink
 - (d) Red and white striped
- Q4.** On selfing pink F1 *Mirabilis* plants, the F2 ratio of Red:Pink:White is:
- (a) 3:1
 - (b) 1:2:1
 - (c) 9:3:3:1
 - (d) 1:3:1
- Q5.** In ABO blood group inheritance, how many alleles of gene "I" exist in the population?
- (a) Two
 - (b) Three
 - (c) Four
 - (d) Six
- Q6.** Who discovered the ABO blood group system?
- (a) Felix Bernstein
 - (b) Karl Landsteiner
 - (c) Carl Correns
 - (d) Gregor Mendel
- Q7.** Which relationship exists between the alleles I-A and I-B in the ABO blood group system?
- (a) I-A is dominant over I-B
 - (b) I-B is dominant over I-A
 - (c) I-A and I-B are co-dominant
 - (d) Both are recessive to i
- Q8.** In co-dominance, the F1 offspring:
- (a) Shows a blended, intermediate phenotype
 - (b) Resembles only the dominant parent
 - (c) Shows both parental phenotypes fully and independently expressed
 - (d) Shows neither parental phenotype
- Q9.** Pleiotropy refers to a situation where:
- (a) A single trait is controlled by multiple genes
 - (b) A single gene influences more than one phenotypic trait

- (c) Two alleles blend to form a new phenotype
- (d) Two genes are linked on the same chromosome

Q10. Polygenic inheritance involves a trait controlled by:

- (a) A single gene
- (b) Two genes
- (c) Three or more genes
- (d) Only genes on the X chromosome

Q11. In Andalusian fowl, a cross between black (BB) and white (bb) birds produces F1 birds that are:

- (a) All black
- (b) All white
- (c) Blue, with fine alternating black/white striped feathers
- (d) Half black, half white

Q12. Selfing the Blue F1 Andalusian fowl gives an F2 ratio of:

- (a) 3 Black : 1 White
- (b) 1 Black : 2 Blue : 1 White
- (c) 9:3:3:1
- (d) 1:1

Q13. A key distinguishing feature of incomplete dominance compared to complete dominance is that:

- (a) The phenotypic ratio equals the genotypic ratio in F2
- (b) F1 always resembles the dominant parent
- (c) Only one allele is needed for full expression
- (d) The trait is always sex-linked

Q14. Which ABO genotype corresponds to blood group "AB"?

- (a) I-A I-A
- (b) I-A I-B
- (c) I-A i
- (d) ii

Q15. A person with genotype I-B i has which blood group phenotype?

- (a) A
- (b) B
- (c) AB
- (d) O

Answer Key & Explanations

- Q1.** (b) Carl Correns — Carl Correns discovered incomplete dominance in 1903 while working on the four o'clock plant, *Mirabilis jalapa*.
- Q2.** (b) Is an intermediate/mixture of both parents, resembling neither parent — In incomplete dominance, F1 shows a blended, intermediate phenotype that resembles neither parent completely, unlike complete dominance or co-dominance.
- Q3.** (c) Pink — RR (red) x rr (white) gives an F1 that is entirely pink (Rr), an intermediate phenotype, illustrating incomplete dominance.
- Q4.** (b) 1:2:1 — Because dominance is incomplete, the F2 phenotypic ratio (1 Red : 2 Pink : 1 White) equals the genotypic ratio (1 RR : 2 Rr : 1 rr).
- Q5.** (b) Three — The ABO blood group gene 'I' has three alleles: I-A, I-B, and i, making it a classic example of multiple allelism.
- Q6.** (b) Karl Landsteiner — Karl Landsteiner discovered the ABO blood group system in 1901; its inheritance pattern was later worked out by Felix Bernstein in 1924.
- Q7.** (c) I-A and I-B are co-dominant — I-A and I-B are co-dominant with each other; both are dominant over the allele i, which produces no sugar antigen.
- Q8.** (c) Shows both parental phenotypes fully and independently expressed — In co-dominance, both alleles are expressed fully and independently in the heterozygote, so F1 resembles both parents, unlike the blended phenotype of incomplete dominance.
- Q9.** (b) A single gene influences more than one phenotypic trait — Pleiotropy is when a single gene produces multiple, seemingly unrelated phenotypic effects, mainly by affecting metabolic pathways.
- Q10.** (c) Three or more genes — Polygenic (multifactorial or quantitative) inheritance involves a trait controlled by three or more genes whose effects are additive.
- Q11.** (c) Blue, with fine alternating black/white striped feathers — BB (black) x bb (white) Andalusian fowl produce an F1 that is entirely Blue (Bb), showing fine alternating black and white striping, an example of incomplete dominance.
- Q12.** (b) 1 Black : 2 Blue : 1 White — Since dominance is incomplete in Andalusian fowl, the F2 ratio is 1 Black : 2 Blue : 1 White, matching the underlying genotypic ratio.
- Q13.** (a) The phenotypic ratio equals the genotypic ratio in F2 — In incomplete dominance the F2 phenotypic ratio equals the genotypic ratio, since heterozygotes have their own distinct, intermediate phenotype.
- Q14.** (b) I-A I-B — The AB blood group arises from the genotype I-A I-B, in which both co-dominant alleles are expressed.
- Q15.** (b) B — Since I-B is dominant over i, the genotype I-B i produces the B blood group phenotype.